



Spatial distribution of genetic variation in a natural beech stand (*Fagus sylvatica* L.) based on microsatellite markers

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Abstract

The spatial distribution of alleles is described in a naturally regenerated, isolated pure beech (*Fagus sylvatica* L.) stand consisting of 99 adult trees. After testing nine microsatellite loci originally developed for *F. crenata*, each tree was genotyped at four well-scorable microsatellite loci. Specific primers were developed for one locus of *F. sylvatica* L. For the characterization of spatial genetic structures, two different statistics were used. One method is based on the mean genetic distance between trees in different spatial distance classes, and the other one is Moran's index *I*. The results show the same tendency of a strong family structure in the distance classes up to 30 m in comparison with that expected for a spatially non-systematic distribution of genotypes. In general, microsatellites are more useful to detect spatial genetic structures than allozymes. Spatial genetic structures are influenced by unpredictable factors such as wind direction at anthesis and can therefore vary from year to year. We recommend that seed collections should cover large areas in order to prevent a preponderance of few families and a reduction of the adaptive potential of the next generation.

Introduction

Forest tree populations consist of sessile, long-lived organisms which must survive temporally varying and spatially heterogeneous environmental conditions that are presently also affected by accelerated global climate change. Hence, the presence and maintenance of genetic variation is important for the long-term existence and stability of forest tree populations. A sufficient genetic variation in populations is a prerequisite for their ability to survive heterogeneous temporal and spatial conditions and to preserve their adaptability for future generations.

European beech (*Fagus sylvatica* L.) is one of the carrier tree species of various forest ecosystems and covers a large continuous geographic range in Europe. This species is monoecious, anemophilous, mainly outcrossing but self-fertile. In a number of

allozyme studies, the genetic structure of beech populations, their mode of reproduction and their genetic diversity in space and time were analyzed (Gregorius et al. 1986; Comps et al. 1990, 1991; Müller-Starck and Ziehe 1991; Müller-Starck and Starke 1993; Merzeau et al. 1994, Starke et al. 1995; Müller-Starck R. 1996; Konnert et al. 2000).

The investigations of Starke and Müller-Starck (1992) on the basis of allozyme analysis showed that the composition of the effective pollen clouds and, therefore, the patterns of outcrossing in beech stands were affected by the spatial distribution of the trees. The development of molecular markers, such as PCR-based nuclear-encoded microsatellite markers (SSRs), now leads to a more efficient analysis of the extent and mechanisms of gene dispersal in angiosperm populations such as beech.

Within naturally regenerating forest tree populations, limited gene flow can induce local genetic

differentiation. Even though forest trees show considerable pollen flow, spatial genetic structures can be recognized on a small scale, e.g. the effective pollen movement in *Quercus alba* is more restricted than previously expected as demonstrated by an effective pollination neighborhood, in the shape of a circle of about 17 m radius (Smouse et al. 2001). Both the maximum and the average transport distances of effective beech pollen have also been found to be rather restricted (Müller-Starck 1996; Wang 2001). The knowledge of the spatial genetic structure within populations has important practical relevance for the conservation and utilization of forest tree genetic resources, since the reduction of population density in the course of natural regeneration can enforce mating preferences for neighboring trees. If neighboring trees in spite of the reduced density on the average still are more closely related than more distant trees, preferential mating among neighbors results in inbreeding (Hattermer et al. 2001), and inbreeding affects populations of predominantly outbreeding species by depression concerning for example reproductive fitness (Frankham 1995). Consequently, the analysis of fine-scale genetic structures is a prerequisite for the development of adequate sampling strategies for conservation purposes.

So far in *Fagus sylvatica* L. dominant (Random Amplified Polymorphic DNA (RAPD)) Welsh and McClelland 1990 and Inter Simple Sequence Repeats (I-SSR) markers were used to estimate certain population genetic parameters (Troggio et al. 1996). Their markers turned out to possess an unexpectedly low efficiency in paternity assignment. Nevertheless, the results were interpreted in accordance with the hypothesis that genetic clustering in beech populations is low. Pastorelli et al. (2003) described the development of microsatellite markers in *F. sylvatica* and *F. orientalis*, but they focussed their study on the techniques of marker development.

Several studies analysed the local and broad-scale genetic structure of different oak and beech species using allozymes and microsatellites (Leonardi and Menozzi 1996; Degen and Scholz 1998; Streiff et al. 1998; Takahashi et al. 2000). In the case of beech species so far, only allozymes were used for the analysis of the spatial distribution of genetic variation. The different species exhibit weak spatial structures of diversity.

In the present study, we adapted microsatellite markers, which were first developed for *F. crenata*

and the closely related species, *F. japonica* (Tanaka et al. 1999), to *F. sylvatica*. Three of these markers show well-scorable bands in *F. sylvatica* and were applied to estimate population genetic parameters and the spatial distribution of the genetic variation within a natural 200 year old stand, which presumably originated from natural regeneration. In a previous study, this stand was also characterized by allozyme analysis (Müller-Starck 1996). Therefore, we are able to compare the effectiveness of the different marker types for sampling strategies in the framework of genetic conservation.

Material and methods

Plant material

Plant material was collected from a small, isolated pure beech stand in the Solling region (Lower Saxony, Germany, Lat. 57° 32', Long. 35° 43', Alt. 370 m). The stand is about 200 years old with little variation in age according to local documents. The stand originated from natural regeneration and was not influenced by silvicultural measures except thinnings. The stand is surrounded by a belt of Norway spruce (*Picea abies*) stands exceeding 500 m in width which was expected to act as a filter for pollen immigration. In fact, a multilocus estimate of the lower bound of the influx of external effective pollen into this stand turned out to be only 2.5% (Müller-Starck, 1996). Still less pollen influx was detected for a different reproduction period by Decarli (2003) some years later. For DNA isolation, buds were harvested from all of the 99 adult trees of the stand.

DNA isolation

The total DNA from buds was extracted using the DNeasy™ Plant Minikit (Qiagen, Hilden, Germany). The amount and the quality of the DNA were analyzed by 0.8% agarose gel electrophoresis with 1 × TAE as running buffer (Sambrook et al. 1989). DNA was stained with ethidium bromide and visualized by UV illumination.

Microsatellite adaptation to *F. sylvatica*

The oligonucleotide primer sequences for *mfc* 5, *mfc* 9 and *mfc* 11 were originally developed for

F. crenata, described and denoted by Tanaka et al. (1999). All tree loci were verified by sequencing. For *mfc* 11, we developed new primers for *F. sylvatica* which are denoted as *mfs* 11 (see Table 1). Microsatellite alleles amplified with the original primer pairs were separated on a 2.0% agarose gel, excised from the gel and purified with the gel extraction kit QIAEX II (Qiagen). The sequencing reaction was carried out with the Big Dye® Terminator v.3.1. Cycle Sequencing Kit (Applied Biosystems) based on the dideoxy-mediated chain termination method (Sanger et al. 1977). Sequencing reactions were run on the ABI 3100 Genetic Analyser (Applied Biosystems). Sequences were analyzed and edited with the sequencing analysis software 3.7 (Applied Biosystems). Sequence alignments were performed with BioEdit version 5.09 (Hall 1999) using ClustalW multiple alignment (Thompson et al. 1994). For locus *mfs* 11, primers were designed with the program Primer 3 from Steve Rozen & Whitehead Institute/MIT Center for Genome Research (http://www-genome.wi.mit.edu/cgi-bin/primer/primer3_www.cgi). Primers were checked for self-annealing, dimer and hairpin formation.

PCR amplification

PCR amplification was performed in a 25 µl volume containing 10 ng template DNA, 10 mM Tris/HCl pH 9.0, 0.2 mM of each dNTP, 1.5 mM MgCl₂, 50 mM KCl, 0.2 µM each of forward and reverse primers and 1 U of *Taq* polymerase (Qiagen, Hot Star Master Mix, Hilden, Germany). All amplifications were performed in a Peltier Thermal Cycler (PTC-0200 version 4.0, MJ Research) with a heated lid under the following conditions: an initial denaturation at 95 °C for 15 min, followed by 30 cycles of 15 s at 94 °C, a 1 min annealing step at 58 °C (primer pairs *mfc* 5 and *mfc* 11), at 55 °C (primer pair *mfs* 11) or 1 min at 53 °C (primer pair *mfc* 9), a 1 min extension step at 72 °C, with a final 30 min at 72 °C. Primer pairs were preliminarily tested by running PCR products on 2% agarose gels in 1 × TAE buffer (see above). Microsatellite alleles were analyzed on an automatic sequencer (ABI 3100 Genetic Analyser). The PCR reactions were carried out with one primer (the forward primer) labeled with the fluorescent dyes 6-FAM (blue) or HEX (green). Fragments were separated on the ABI Genetic Analyser in a multiplex anal-

ysis. GS 500 ROX (fluorescent dye ROX) was used as internal standard (Applied Biosystems). The microsatellite alleles were recognized using the software packages Genescan 3.7 and Genotyper 3.7 from Applied Biosystems.

Statistical data analysis

Genetic parameters

Microsatellite loci were scored for the analysis of the population genetic parameters by using the computer programs GSED (Gillet 1994) and GENALEX (Peakall and Smouse 2001). For each locus, the observed heterozygosity (H_o) and the expected heterozygosity (H_e) were analyzed following the formulae of Hartl and Clark (1997). The effective number of alleles $v = (\sum p_i^2)^{-1}$ was calculated according to Crow and Kimura (1970) and the fixation index

$$F = 1 - \frac{H_o}{H_e} \text{ was calculated according to Wright (1965).} \quad (1)$$

Spatial genetic structure

Two different statistical methods were used to analyze the correlation between the spatial distribution and the genetic variation within this stand. The method described by Degen et al. (2001) is based on the mean genetic distance between pairs of individuals of the same distance class. The second method based on Moran's I analyzes the correlation of the allelic information between pairs of individuals of the same distance class.

The distograms for genetic distance-based measures were produced by the program Spatial Genetic Software (SGS) Version 1.0c (Degen et al. 2001). In this program, the mean genetic distance is calculated according to Gregorius (1978).

For the computation, the genetic data were coded such that single trees received allele frequency values of 0, 0.5 or 1 for each allele at every locus. All alleles were included in the estimation of Moran's I (Moran 1950; Streiff et al. 1998) using the program R 1.6.2 Language and Environment with package "spdep" (<http://www.r-project.org/>) which was adapted by J.R.G. Kleinschmit. In addition, Moran's I was estimated by excluding

Table 1. Characterization of microsatellite markers in *Fagus sylvatica*

Locus	Core sequences	PCR primer (5'–3')	Annealing temperature (°C)	Allele size range (bp)	No alleles	Sequenced allele size
<i>mfc</i> 2	(CT)27	ACAACCTAATCCCACCACTCC GGCTAAAGCGACAACTCTA	60	Multi-locus		
<i>mfc</i> 3	(ATT) 9	TAATGGGATGGAGGTGGAT GGAAGACAAACAATGGAAG	53	No amplific.		
<i>mfc</i> 4	(CT) 6	ATMAGCAACTGAATAAAGA GTCAAGAGAGGGGAAAGAGGT	50	Multi-locus		
<i>mfc</i> 5	(AG) 10	ACTGGGACAAAAAACAAAA GAAGGACCAAGGCACATAAA	58	277–329	21	282,300,321
<i>mfc</i> 7	(GA) 9	AAAATACACTGCCCAAAA CAGGTTTTGGTTTCTTACAC	58	Multi-locus		
<i>mfc</i> 9-2	(GA) 19	TTCCCTCCTCTTCTCAAAT TTATACTTCTCTCTCATCCC	53	182–200	5	182, 196, 199
<i>mfc</i> 11	(AG) 10	ACAGATAAAACAGAAAGCCA TTTGGTTTGTGAGTTTAG	58	312–331	8	319,325,327
<i>mfs</i> 11	(AG) 10	GGGGTGGTTTCAAGTTTC GAAGCCAATTATCACACCAAAAC	55	125–144	9	
<i>mfc</i> 12	(AGG) (AG)6(AG- G)(AG)7(AGG)	ACACCTCACAAATCCACGAA	56	No amplific.		
<i>mfc</i> 13	(CT) 4(GT)2(CA)2	CCCAATAACTAAGAATACCA GAGAGCAGGATGGGATGAAA ACAGCACCTCTCCTCTCTT	55	No amplific.		

the rare alleles, which were represented only once or twice in the samples. Moran's I was calculated according to the following equation:

$$I = \frac{n \sum_{i=1}^n \sum_{j \neq i}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{W \sum_{i=1}^n (x_i - \bar{x})^2}, \quad (2)$$

where $W = \sum_{i=1}^n \sum_{j \neq i}^n w_{ij}$ and n is the number of individuals sampled, x_i is the allele frequency of individual i (1 if individual i is homozygous for a given allele, 0.5 if the individual is heterozygous for the allele and 0 if the individual has no copy of it), $\bar{x}$ is the mean value of x_i for all individuals in the sample, w_{ij} is 1 when the distance between individuals i and j belong to the distance class under investigation, and zero otherwise.

For both procedures, the statistical significance of all measures is evaluated by permutation tests in order to determine deviations from spatially random distributions. The 95% envelope for the correlogram and distogram is obtained after 1,000 permutations under the hypothesis of no spatial correlation.

Results and discussion

Microsatellites were surveyed to characterize the spatial distribution of genetic variation in a naturally regenerated isolated beech stand. The variability at neutral, non-coding microsatellite loci is due to differences in the number of repeat units. This is easily detectable as variation in the lengths of DNA fragments obtained through PCR amplification. Microsatellite primers developed in *F. crenata* were applied for the analysis of *F. sylvatica*. Among the nine tested loci, only three

showed well-scorable polymorphic amplification products with different numbers of alleles in *F. sylvatica*. The loci *mfc* 2, 4 and 7 present a multiple-locus banding pattern, but the primer pairs of loci *mfc* 3, 12 and 13 generated no amplification product in *F. sylvatica* (Table 1).

Twenty one alleles were found for the most variable locus *mfc* 5. This locus was also one of the most variable loci in *F. crenata* (16 alleles) and in *F. japonica* (18 alleles) (Tanaka et al. 1999). Eighteen alleles were detected among 70 individuals of *F. sylvatica* in an Italian natural population and 15 alleles among 45 individuals of three different populations in the closely related species of *F. orientalis* (Pastorelli et al. 2003). The newly developed locus *mfs* 11 possesses nine alleles and provides for a precise identification of the heterozygous individuals. With the primer pairs of locus *mfc* 11, which shows 10 and 4 alleles in *F. crenata* and *F. japonica*, respectively (Tanaka et al. 1999), eight alleles were detected at our study site. Among the analyzed trees, 54 individuals were scored as homozygous. Twenty of these individuals were reclassified as heterozygotes with the new primer pair *mfs* 11 resulting in increased values for observed heterozygosity (H_o) (0.625 versus 0.417) as well as expected (0.652 versus 0.574) heterozygosity (H_e), and a great discrepancy for the fixation indices. Although the fixation index of locus *mfc* 11 supports a population substructure (Hartl and Clark 1997), the findings on locus *mfs* 11 did not corroborate this hypothesis (Table 2).

The locus *mfc* 9-2 presents five alleles and a large proportion of putatively homozygous genotypes suggesting the presence of a null-allele. The presence of null-alleles of microsatellite regions was also described for two other loci in *F. sylvatica* and *F. japonica* developed by Pastorelli et al.

Table 2. Variation parameters of microsatellite loci analyzed in *F. sylvatica*

Locus	Number of alleles encountered	v_2	H_o	H_e	F
<i>mfc</i> 5	21	9.64	0.663	0.899	0.262
<i>mfc</i> 9-2	5	3.93	0.427	0.745	0.427
<i>mfc</i> 11	8	2.35	0.417	0.574	0.274
<i>mfs</i> 11	9	2.87	0.625	0.652	0.041
Mean	10.75	4.25	0.572	0.765	0.192

Means were based on *mfc* 5, *mfc* 9-2 and *mfs* 11 only. v_2 describes the genetic diversity or the effective number of alleles, H_o and H_e the observed and expected heterozygosity and F the fixation index.

(2003) and Tanaka et al. (1999), respectively. This was also indicated by a homozygote excess.

In the case of *mfs* 11, the extent of null-alleles could obviously be diminished by special primer design. Figure 2 shows the spatial distribution of the most frequent alleles of locus *mfc* 5 and *mfs* 11 within the study site. Although locus *mfc* 5 is characterized by 21 alleles and a high genetic diversity (Table 2), a strong family structure of neighboring trees is clearly reflected in the distribution of the most frequent allele of 302 bp length. This allele is not only carried by seven trees in homozygous condition, but these trees are concentrated as a group at the northern edge of the stand. This strong homozygosity for this allele also contributes to explain the deviation from the Hardy–Weinberg structure at this locus as indicated by the fixation index of this locus (Table 2). However, in contrast to this concentrated distribution of the allele of 302 bp, the allele of 300 bp at locus *mfc* 5 with nearly the same frequency shows a more homogeneous distribution (Figure 1a).

The spatial distribution of the different genotypes of locus *mfs* 11 confirms this result which also indicates that groups of nearby individuals are related (Figure 1b). This effect of clustering could be expected due to a limited gene flow by seed dispersal and preferential mating among possibly related neighbor individuals if sufficient selection with inbreeding depression does not occur. The allozyme analysis of this stand showed that the migration distance of both pollen and seeds within the stand is limited to approximately 50 m (Müller-Starck R. 1996) promoting the effect of clustering of related genotypes and the mating among related individuals. When working in the very same stand, Wang (2001) found that the average transport distance of both pollen and seeds is much lower. The seeds migrated only about half the distance of the pollen, which is eventually crucial for the origin of spatial genetic structures in this species. Similar results were obtained in two other beech stands by also utilizing allozyme loci (Doúnavi 2000).

The joint spatial autocorrelation demonstrated by Moran's *I* (Figure 2a) using allelic doses of the individuals at loci *mfc* 5 and *mfs* 11 shows the tendency of a strong family structure in the distance classes until 30 m. The autocorrelation found by allozyme analysis showed only a weak spatial clustering within this distance class in this

stand. The strong correlation for the first two spatial distance classes is supported by the distogram in Figure 2b. Trees at a distance of up to 30 m are genetically more similar than expected under the hypothesis of a spatially random distribution of genotypes. The analysis was based only on the loci *mfc* 5 and *mfs* 11, since the inclusion of loci *mfc* 9-2 and *mfc* 11 would lead to an overestimation of this effect due to the large proportion of individuals scored erroneously as homozygous but putatively carrying a null-allele. The exclusion of the rare alleles at the loci *mfc* 5 and *mfs* 11 which were represented only once or twice in our samples had no significant effect on our results.

Few studies on forest trees have demonstrated significant spatial structuring. Streiff et al. (1998) found similar results in *Quercus robur* and *Q. petraea* by using microsatellite and isozyme markers. They found a significant, but weak, spatial genetic structure by using several markers and different statistics. They explained their results of extremely weak spatial structure, which is in contrast to our results, by elimination of inbred individuals resulting from matings between related individuals. Ueno et al. (2000) found in an old-growth evergreen forest of *Camellia japonica* also a positive autocorrelation over short-distance classes (up to 50 m) which was explained by limited seed dispersal and microsite heterogeneity of the stand.

However, in the case of limited seed and pollen dispersal also in small populations with a limited number of trees, microsatellites with their generally larger amount of variation exhibit stronger spatial structures than allozymes. A small number of significant autocorrelation values were found in beech based on allozyme analysis (Merzeau et al. 1994). Only one adult forest showed a significant positive autocorrelation within patches. Based on data of 14 beech populations, Leonardi and Menozzi (1996) computed the autocorrelation as a standard normal deviate (SND) which is the difference between the frequency of observed and expected genotype pairs in each distance class standardized for the standard deviation of the respective class. Their results parallel our observations from microsatellite data of a significant autocorrelation within a distance of up to 30 m. Additional stands which were either naturally regenerated or planted have to be analyzed in order to enable us to generalize the present results.

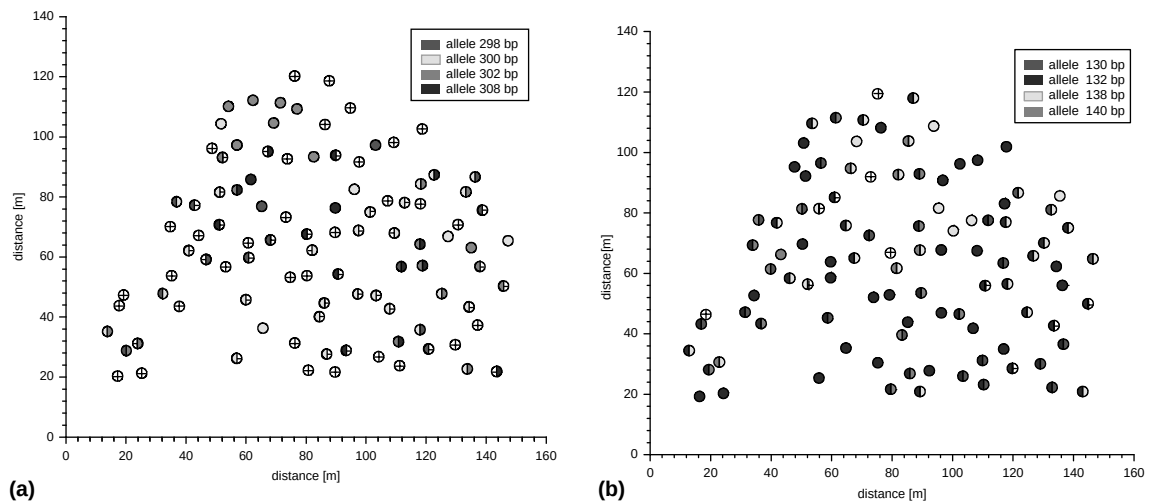


Figure 1. Spatial distribution of the most frequent alleles of locus *mfc* 5 and *mfs* 11 within the study site. Less frequent alleles are denoted by white symbols: (a) Gene locus *mfc* 5, (b) Gene locus *mfs* 11.

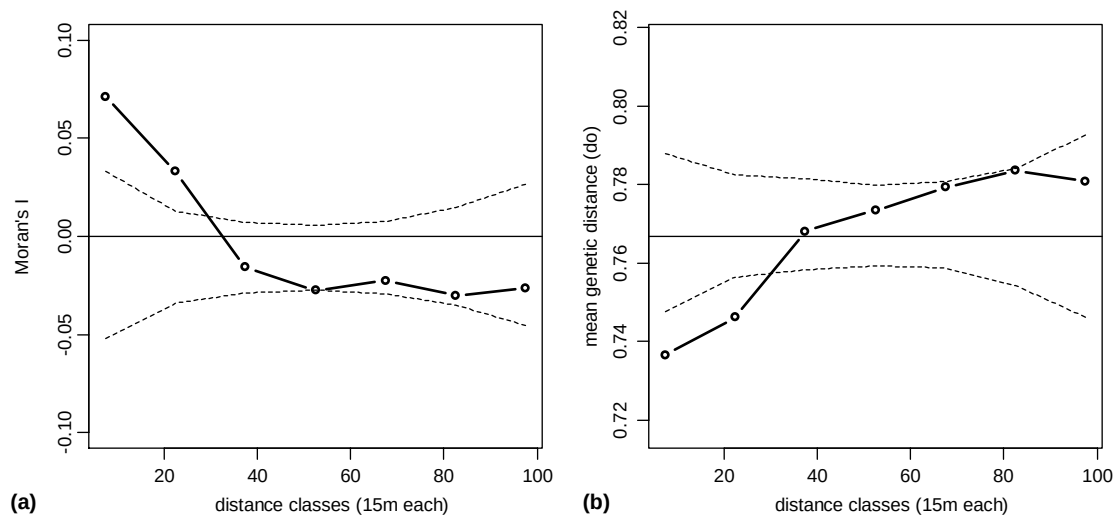


Figure 2. Joint spatial autocorrelation at the loci *mfc* 5 and *mfs* 11. Values were calculated over all alleles. Thin dotted lines show the 95% envelope under the null hypothesis of no differentiation, which was obtained after 1,000 permutations of the multilocus genotypes: (a) Correlogram using Moran's index I , (b) Distogram using mean distance d_0 (Gregorius 1978).

Conclusions

Since efforts devoted to the genetic conservation of beech populations concentrate on measures *in situ*, the reproduction system of beech populations has to be better understood. In order to be genetically effective, silvicultural operations in the framework of natural regeneration have to be in accordance with the genetic processes involved in reproduction, subsequent seed dispersal and stand treatment.

For instance, the findings on genotypic frequency structure summarized in Table 2 can hardly be explained under the hypothesis of random mating in the previous generation. Under this hypothesis, fixation indices of varying sizes in this order of magnitude have conventionally been interpreted by the existence of null-alleles occurring with moderate but varying frequencies. Since null-alleles simulate inbreeding structures, loci *mfc* 9-2 and *mfc* 11 with suspected null-alleles were omitted

from the analysis of spatial structure. The hypothesis of inbreeding in the absence of null-alleles would contradict all results that have been derived by the use of allozymes in the past. Even in the very stand that was analyzed in the present study, Müller-Starck (1996) found fixation indices at 10 allozyme loci averaging 0.04 in the adult trees, 0.07 in the seeds and 0.02 in young seedlings. These parameter estimates would indicate a mating system close to randomness rather than to inbreeding.

However, the findings on spatial structures can hardly be explained by random dispersal of genetic variants by both pollen and seed. This was shown by the use of allozymes in the cited literature. Even the relatively faint clumping pattern of genotypes that can be observed at allozyme loci (Doúnavi 2000) is diminished but not completely removed by thinning operations (Doúnavi et al. 2002).

Since the effective pollen of beech is dispersed over restricted distances and the species is also predominantly barochorous (Wang 2001), a clumped spatial pattern of genetic variants had to be expected in this stand and other stands of the type studied. One has to envisage how a prevailing southerly wind direction during anthesis of beech would affect the representation of an allele with the same distribution as the one with 302 bp of *mfc* 5 in the subsequent generation. As Decarli (2003) has shown in the stand studied in the present project, the influx of some external alleles (16 out of > 6,000 or fractions of a percent) also did show a spatial pattern in a particular year. Since the same must be postulated for the alleles occurring among the adult trees within the stand, the wind conditions during the flowering period must have a bearing on the change of genetic structures during reproduction. We may derive that under the given conditions of spatial structure, it requires much silvicultural skill on the basis of careful genetic planning to safeguard the representation of all genetic variants in the subsequent generation without undue man-induced change of frequency structures.

Also, for the practice of commercial seed collection in stands with a low density of trees, our results show that it is very important to attain knowledge not only of the genetic composition but also of the spatial genetic structure of natural beech populations. The majority of beech stands that are approved for seed collection is natural. There exists no legal requirements how seed col-

lections have to be carried out. Therefore, seed collections are done more or less arbitrarily. We also recommend that the sampling for conservation purposes should cover larger areas in order to prevent a reduction of genetic diversity and, hence, of the adaptive potential of the next generation. Since genetic structures of seeds can be influenced by many factors, for example wind direction and fructification intensity, it would be useful to compare seed collections sampled in different years.

Morgante et al. (2002) found that microsatellites are abundant in transcribed and other single-copy regions of plant genomes. In microbial genomes, microsatellites are over-represented in stress response genes (Rocha et al. 2002). These findings indicate that microsatellites might be useful markers not only for the determination of spatial genetic structures but also for the estimation and the monitoring of the adaptive capacity of populations. There exist research needs in the area of how to implement information on spatial genetic structures into efficient measures of the conservation of genetic resources of beech *in situ*.

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